

Práctica 2

Redes Neuronales

Curso 2024/2025

Máster Universitario en Inteligencia Artificial, Reconocimiento
de Formas e Imagen Digital

Departamento de Sistemas Informáticos y Computación

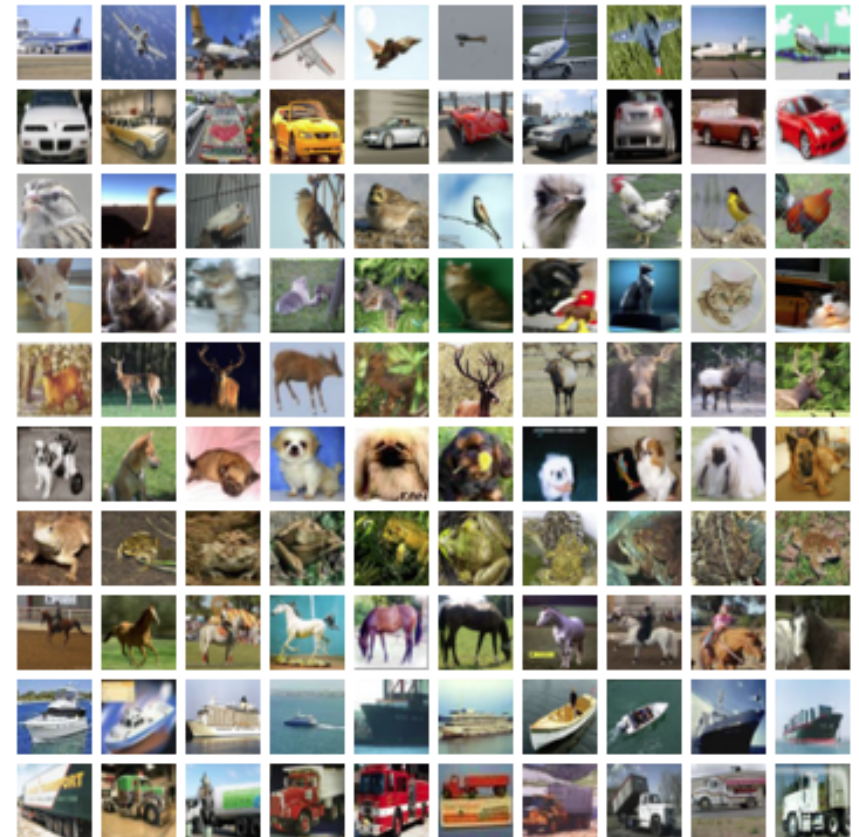
Goals

- Implement generalization and performance improvements:
 - Drop-out
 - Batch norm
 - Noise regularization
 - Learning rate annealing
 - ...
- Improvements just tested over MNIST using Fully Connected (FC)
- **Implement convolutional networks (CNNs)**
- **Test these improvements over CIFAR-10 using CNNs**

CIFAR-10

- Images of 10 different classes
- Size 32x32 pixels
- 50K training, 10K test

airplane
automobile
bird
cat
deer
dog
frog
horse
ship
truck



CIFAR Convolutional Nets

- Use CNNs
- Improvements:
 - Dropout
 - Batch norm
 - Noise regularization
 - Learning rate schedulers
 - Deeper CNNs
 - Data augmentation
 - Residual connections

Lab Evaluation

- CIFAR up to 2 points:

Dataset	Error	Points
CIFAR Conv	14%	0.5
CIFAR Conv	12%	0.5
CIFAR Conv	10%	0.5
CIFAR Conv	8%	0.5